

0.1 递推法与数学归纳法

命题 0.1 (三对角行列式)

求下列行列式的递推关系式 (空白处均为 0):

$$D_n = \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-1} & b_{n-1} \\ & & & & c_{n-1} & a_n \end{vmatrix} = a_n D_{n-1} - b_{n-1} c_{n-1} D_{n-2} (n \geq 2).$$

笔记 Show/Hide Note

记忆三对角行列式的计算方法和结果: $D_n = a_n D_{n-1} - b_{n-1} c_{n-1} D_{n-2} (n \geq 2)$,

即按最后一列 (或行) 展开得到递推公式.

解 Show/Hide Solution

显然 $D_0 = 1, D_1 = a_1$. 当 $n \geq 2$ 时, 我们有

$$\begin{aligned} D_n &= \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-1} & b_{n-1} \\ & & & & c_{n-1} & a_n \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-2} & b_{n-2} \\ & & & & c_{n-2} & a_{n-1} & b_{n-1} \\ & & & & & c_{n-1} & a_n \end{vmatrix} \\ &\quad \underline{\underline{\text{按最后一列展开}}} a_n \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-2} & b_{n-2} \\ & & & & c_{n-2} & a_{n-1} \end{vmatrix} - b_{n-1} \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-3} & b_{n-3} \\ & & & & c_{n-3} & a_{n-2} & b_{n-2} \\ & & & & & 0 & c_{n-1} \end{vmatrix} \\ &\quad \underline{\underline{\text{第二项按最后展开}}} a_n \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-2} & b_{n-2} \\ & & & & c_{n-2} & a_{n-1} \end{vmatrix} - b_{n-1} c_{n-1} \begin{vmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-3} & b_{n-3} \\ & & & & c_{n-3} & a_{n-2} \end{vmatrix} \\ &= a_n D_{n-1} - b_{n-1} c_{n-1} D_{n-2}. \end{aligned}$$

□

推论 0.1 (三对角矩阵的行列式)

计算 n 阶行列式 ($bc \neq 0$):

$$D_n = \begin{vmatrix} a & b & & & \\ c & a & b & & \\ & c & a & b & \\ & & \ddots & \ddots & \ddots \\ & & & c & a & b \\ & & & & c & a \end{vmatrix} = \begin{cases} \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}, a^2 \neq 4bc, \\ (n+1) \left(\frac{a}{2}\right)^n, a^2 = 4bc. \end{cases},$$

其中 $\alpha = \frac{a + \sqrt{a^2 - 4bc}}{2}, \beta = \frac{a - \sqrt{a^2 - 4bc}}{2}$.



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解递推式: $D_n = aD_{n-1} - bcD_{n-2} (n \geq 2)$ 对应的特征方程: $x^2 - ax + bc = 0$ 得到两根 $\alpha = \frac{a + \sqrt{a^2 - 4bc}}{2}, \beta = \frac{a - \sqrt{a^2 - 4bc}}{2}$, 由 Vieta 定理可知 $a = \alpha + \beta, bc = \alpha\beta$.

解 Show/Hide Solution

由命题 0.1 可知, 递推式为 $D_n = aD_{n-1} - bcD_{n-2} (n \geq 2)$. 又易知 $D_0 = 1, D_1 = a$. 令 $\alpha = \frac{a + \sqrt{a^2 - 4bc}}{2}, \beta = \frac{a - \sqrt{a^2 - 4bc}}{2}$, 则 $a = \alpha + \beta, bc = \alpha\beta$, 于是 $D_n = (\alpha + \beta)D_{n-1} - \alpha\beta D_{n-2} (n \geq 2)$. 从而

$$D_n - \alpha D_{n-1} = \beta (D_{n-1} - \alpha D_{n-2}), D_n - \beta D_{n-1} = \alpha (D_{n-1} - \beta D_{n-2}).$$

于是

$$\begin{aligned} D_n - \alpha D_{n-1} &= \beta^{n-1} (D_1 - \alpha D_0) = \beta^{n-1} (a - \alpha) = \beta^n, \\ D_n - \beta D_{n-1} &= \alpha^{n-1} (D_1 - \beta D_0) = \alpha^{n-1} (a - \beta) = \alpha^n. \end{aligned}$$

因此, 若 $a^2 \neq 4bc$ (即 $\alpha \neq \beta$), 则联立上面两式, 解得

$$D_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta};$$

若 $a^2 = 4bc$ (即 $\alpha = \beta$), 则由 $a = \alpha + \beta$ 可知, $\alpha = \beta = \frac{a}{2}$. 又由 $D_n - \alpha D_{n-1} = \beta^n$ 可得

$$D_n = \left(\frac{a}{2}\right)^n + \frac{a}{2} D_{n-1} = \left(\frac{a}{2}\right)^n + \frac{a}{2} \left(\left(\frac{a}{2}\right)^{n-1} + \frac{a}{2} D_{n-2} \right) = 2 \left(\frac{a}{2}\right)^n + \left(\frac{a}{2}\right)^2 D_{n-2} = \cdots = n \left(\frac{a}{2}\right)^n + \left(\frac{a}{2}\right)^n D_0 = (n+1) \left(\frac{a}{2}\right)^n.$$

综上, 我们有

$$D_n = \begin{cases} \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}, a^2 \neq 4bc, \\ (n+1) \left(\frac{a}{2}\right)^n, a^2 = 4bc. \end{cases}$$

□

命题 0.2 (三对角矩阵的特征值)

设 a, b, c 为复数且 $bc \neq 0$, 证明下列 n 阶矩阵 A 可对角化

$$A = \begin{pmatrix} a & b & & & \\ c & a & b & & \\ & c & a & b & \\ & & \ddots & \ddots & \ddots \\ & & & c & a & b \\ & & & & c & a \end{pmatrix}$$

并且矩阵 A 的全体特征值为

$$\lambda = a + 2\sqrt{bc} \cos \frac{k\pi}{n+1}, 1 \leq k \leq n.$$

证明 Show/Hide Proof

证法一: 我们先来计算 A 的特征多项式 $|\lambda I_n - A|$. 设 x_1, x_2 是二次方程 $x^2 - (\lambda - a)x + bc = 0$ 的两个根, 则当 $(\lambda - a)^2 = 4bc$ 时, 由 **推论 0.1** 可得

$$|\lambda I_n - A| = (n+1) \left(\frac{\lambda - a}{2} \right)^n = (n+1)(2bc)^n \neq 0.$$

此时 A 的特征多项式无解.

当 $(\lambda - a)^2 \neq 4bc$ 时, 由 **推论 0.1** 可得

$$|\lambda I_n - A| = \frac{x_1^{n+1} - x_2^{n+1}}{x_1 - x_2}.$$

注意到 x_1, x_2 都是关于 λ 的连续函数, 要求 A 的特征值 λ , 即是求 λ 的值, 使得 $|\lambda I_n - A| = 0$, 而这也等价于 $x_1^{n+1} = x_2^{n+1}$. 令 $\omega = \cos \frac{2\pi}{n+1} + i \sin \frac{2\pi}{n+1}$ 为 1 的 $n+1$ 次方根, 则由 $x_1^{n+1} = x_2^{n+1}$ 可得 $x_1 = x_2 \omega^k (1 \leq k \leq n)$. 由 Vieta 定理可得 $x_1 x_2 = bc$, 在选定 bc 的某一平方根 $\sqrt{bc}$ 之后, 可解出

$$x_1 = \sqrt{bc} \left(\cos \frac{k\pi}{n+1} + i \sin \frac{k\pi}{n+1} \right), x_2 = \sqrt{bc} \left(\cos \frac{k\pi}{n+1} - i \sin \frac{k\pi}{n+1} \right), 1 \leq k \leq n.$$

再次由 Vieta 定理可得 $\lambda - a = x_1 + x_2 = 2\sqrt{bc} \cos \frac{k\pi}{n+1}$, 即

$$\lambda = a + 2\sqrt{bc} \cos \frac{k\pi}{n+1}, 1 \leq k \leq n.$$

容易验证上述 n 个数确实是 A 的 n 个不同的特征值, 从而 A 可对角化.

证法二: 简记三对角矩阵 $A = T(a, b, c)$, 要证 A 可对角化, 只要证 $A - aI_n$ 可对角化即可, 故不妨设 $a = 0$. 由于 $bc \neq 0$, 故命题??的方法, 依次将第 $i+1$ 行乘以 $\sqrt{(\frac{b}{c})^i}$, 再将第 $i+1$ 列乘以 $\sqrt{(\frac{c}{b})^i} (1 \leq i \leq n-1)$, 则可得到 A 复相似于三对角矩阵 $T(0, \sqrt{bc}, \sqrt{bc}) = \sqrt{bc} \cdot T(0, 1, 1)$. 因为三对角矩阵 $T(0, 1, 1)$ 是实对称矩阵, 故 $\sqrt{bc} \cdot T(0, 1, 1)$ 可对角化, 从而 A 也可对角化. □

例题 0.1 设 n 阶三对角阵

$$A = \begin{pmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & a_{n-1} & b_{n-1} \\ & & & & c_{n-1} & a_n \end{pmatrix},$$

$$\mathbf{B} = \begin{pmatrix} a_1 & b_1 c_1 & & & \\ 1 & a_2 & b_2 c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & a_{n-1} & b_{n-1} c_{n-1} \\ & & & 1 & a_n \end{pmatrix}.$$

设矩阵 $\mathbf{A}$ 的特征值为 $\lambda_1, \lambda_2, \dots, \lambda_n$. 求矩阵 $\mathbf{B}$ 的特征值.

解 [Show/Hide Solution](#)

矩阵 $\mathbf{A}$ 和 $\mathbf{B}$ 的特征多项式分别为

$$f_{\mathbf{A}}^{(n)}(\lambda) = \begin{vmatrix} \lambda - a_1 & -b_1 & & & \\ -c_1 & \lambda - a_2 & -b_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & -b_{n-1} \\ & & & -c_{n-1} & \lambda - a_n \end{vmatrix},$$

$$f_{\mathbf{B}}^{(n)}(\lambda) = \begin{vmatrix} \lambda - a_1 & -b_1 c_1 & & & \\ -1 & \lambda - a_2 & -b_2 c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & \ddots & \lambda - a_{n-1} & -b_{n-1} c_{n-1} \\ & & & & -1 & \lambda - a_n \end{vmatrix}.$$

当 $n = 1$ 时,

$$f_{\mathbf{A}}^{(1)}(\lambda) = \lambda - a_1 = f_{\mathbf{B}}^{(1)}(\lambda).$$

当 $n = 2$ 时,

$$f_{\mathbf{A}}^{(2)}(\lambda) = \begin{vmatrix} \lambda - a_1 & -b_1 \\ -c_1 & \lambda - a_2 \end{vmatrix} = (\lambda - a_1)(\lambda - a_2) - b_1 c_1,$$

$$f_{\mathbf{B}}^{(2)}(\lambda) = \begin{vmatrix} \lambda - a_1 & -b_1 c_1 \\ -1 & \lambda - a_2 \end{vmatrix} = (\lambda - a_1)(\lambda - a_2) - b_1 c_1.$$

故 $f_{\mathbf{A}}^{(2)}(\lambda) = f_{\mathbf{B}}^{(2)}(\lambda)$.

设 $n > 2$, 则由命题 0.1 有

$$f_{\mathbf{A}}^{(n)}(\lambda) = (\lambda - a_n) f_{\mathbf{A}}^{(n-1)}(\lambda) - b_{n-1} c_{n-1} f_{\mathbf{A}}^{(n-2)}(\lambda),$$

$$f_{\mathbf{B}}^{(n)}(\lambda) = (\lambda - a_n) f_{\mathbf{B}}^{(n-1)}(\lambda) - b_{n-1} c_{n-1} f_{\mathbf{B}}^{(n-2)}(\lambda).$$

由于 $f_{\mathbf{A}}^{(1)}(\lambda) = f_{\mathbf{B}}^{(1)}(\lambda)$, $f_{\mathbf{A}}^{(2)}(\lambda) = f_{\mathbf{B}}^{(2)}(\lambda)$, 而 $f_{\mathbf{A}}^{(n)}(\lambda)$ 与 $f_{\mathbf{B}}^{(n)}(\lambda)$ 有相同的递推式, 所以对任意正整数 n , $f_{\mathbf{A}}^{(n)}(\lambda) = f_{\mathbf{B}}^{(n)}(\lambda)$, 从而矩阵 $\mathbf{B}$ 的特征值为 $\lambda_1, \lambda_2, \dots, \lambda_n$.

□

 练习 0.1 求证: n 阶行列式

$$|A| = \begin{vmatrix} \cos x & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 2\cos x & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 2\cos x & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 2\cos x & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 2\cos x \end{vmatrix} = \cos nx.$$

解 Show/Hide Solution

解法一:

设 $|A| = D_n$, 其中 n 表示 $|A|$ 的阶数 ($n \geq 0$). 易知 $D_0 = 1, D_1 = \cos x$.

从而 $|A| = D_n \xrightarrow{\text{按最后一列展开}} 2\cos x D_{n-1} - D_{n-2} (n \geq 2)$.

命题 0.1

其对应的特征方程为 $\lambda^2 = 2\cos x \lambda - 1$, 解得 $\lambda_1 = \cos x + i \sin x, \lambda_2 = \cos x - i \sin x$.

于是当 $n \geq 2$ 时, 我们有 $D_n = (\lambda_1 + \lambda_2) D_{n-1} + \lambda_1 \lambda_2 D_{n-2}$.

进而

$$\begin{aligned} D_n - \lambda_1 D_{n-1} &= \lambda_2 (D_n - \lambda_1 D_{n-1}), \\ D_n - \lambda_2 D_{n-1} &= \lambda_1 (D_n - \lambda_2 D_{n-1}). \end{aligned} \quad (1)$$

由此可得

$$\begin{aligned} D_n - \lambda_1 D_{n-1} &= \lambda_2^{n-1} (D_1 - \lambda_1 D_0) = -i \sin x \cdot \lambda_2^{n-1}, \\ D_n - \lambda_2 D_{n-1} &= \lambda_1^{n-1} (D_1 - \lambda_2 D_0) = i \sin x \cdot \lambda_1^{n-1}. \end{aligned}$$

若 $x \neq k\pi (k \in \mathbb{Z})$, 则联立上面两式, 解得

$$\begin{aligned} D_n &= \frac{i \sin x \cdot \lambda_1^n + i \sin x \cdot \lambda_2^n}{\lambda_1 - \lambda_2} = \frac{i \sin x \cdot (\cos x + i \sin x)^n + i \sin x \cdot (\cos x - i \sin x)^n}{2i \sin x} \\ &\xrightarrow{\text{Euler 公式}} \frac{i \sin x \cdot e^{nxi} + i \sin x \cdot e^{-nxi}}{2i \sin x} = \frac{i \sin x \cdot (\cos nx + i \sin nx) + i \sin x \cdot (\cos nx - i \sin nx)}{2i \sin x} \\ &= \frac{2i \sin x \cdot \cos nx}{2i \sin x} = \cos nx. \end{aligned}$$

若 $x = k\pi (k \in \mathbb{Z})$, 则 $\lambda_1 = \lambda_2 = \cos k\pi$. 从而由(1)式可得, $D_n - \cos k\pi D_{n-1} = -i \sin x \cdot (\cos k\pi) = 0$.

于是

$$D_n = \cos k\pi D_{n-1} = (\cos k\pi)^2 D_{n-2} = \cdots = (\cos k\pi)^n D_0 = (\cos k\pi)^n = (-1)^{kn} = \cos(nk\pi) = \cos nx.$$

解法二: 仿照练习 0.3 中的数学归纳法证明.

□

 练习 0.2 求下列 n 阶行列式的值:

$$D_n = \begin{vmatrix} 1-a_1 & a_2 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{vmatrix}.$$

 笔记 Show/Hide Note

观察原行列式我们可以得到, D_n 的每列和有一定的规律, 即除了第一列和最后一列, 中间每列和均为 0. 并且 D_n 是三对角行列式. 因此, 我们既可以应用三对角行列式的结论 (即命题 0.1), 又可以使用求和法进行求解. 如果我们直接应用三对角行列式的结论 (即命题 0.1), 按照对一般的三对角行列式展开的方法能得到相应递推式, 但是

这样得到的递推式并不是相邻两项之间的递推, 后续求解通项并不简便. 又因为使用求和法计算行列式后续计算一般比较简便所以我们先采用求和法进行尝试.

解 Show/Hide Solution

解法一: 当 $n \geq 1$ 时, 我们有

$$D_n = \begin{vmatrix} 1-a_1 & a_2 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{vmatrix} \xrightarrow[r_i+r_1]{i=2,\dots,n} \begin{vmatrix} -a_1 & 0 & 0 & 0 & \cdots & 0 & 1 \\ -1 & 1-a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{vmatrix}$$

$$\xrightarrow{\text{按第一行展开}} -a_1 D_{n-1} + (-1)^{n+1} \begin{vmatrix} -1 & 1-a_2 & a_3 & 0 & \cdots & 0 \\ 0 & -1 & 1-a_3 & a_4 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 \end{vmatrix}$$

$$= -a_1 D_{n-1} + (-1)^{n+1} (-1)^{n-1}$$

$$= 1 - a_1 D_{n-1}.$$

其中 D_{n-i} 表示 D_{n-i+1} 去掉第一行和第一列得到的 $n-i$ 阶行列式, $i = 1, 2, \dots, n-1$. (或者称 D_{n-i} 表示以 $a_{i+1}, \dots, a_n$ 为未定元的 $n-i$ 阶行列式, $i = 1, 2, \dots, n-1$)

由递推不难得到

$$D_n = 1 - a_1 (1 - a_2 D_{n-2}) = 1 - a_1 + a_1 a_2 D_{n-2} = \cdots = 1 - a_1 + a_1 a_2 - a_1 a_2 a_3 + \cdots + (-1)^n a_1 a_2 \cdots a_n.$$

解法二: 仿照练习0.3中的数学归纳法证明.

□

命题 0.3

计算 n 阶行列式:

$$D_n = \begin{vmatrix} x_1 & y & y & \cdots & y & y \\ z & x_2 & y & \cdots & y & y \\ z & z & x_3 & \cdots & y & y \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ z & z & z & \cdots & x_{n-1} & y \\ z & z & z & \cdots & z & x_n \end{vmatrix}.$$

 **笔记** Show/Hide Note

解法二: $f(x) \triangleq \begin{vmatrix} x_1+x & y+x & \cdots & y+x \\ z+x & x_2+x & \cdots & y+x \\ \vdots & \vdots & & \vdots \\ z+x & z+x & \cdots & x_n+x \end{vmatrix} = \begin{vmatrix} x_1+x & y+x & \cdots & y+x \\ z-x_1 & x_2-y & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ z-x_1 & z-y & \cdots & x_n-y \end{vmatrix}$, 再按第一行展开可得 $f(x)$ 一定为关于 x 的线性函数.

解 Show/Hide Solution

解法一(小拆分法): 对第 n 列进行拆分即可得到递推式: (对第 1 或 n 行(或列) 拆分都可以得到相同结果)

$$\begin{aligned}
 D_n &= \begin{vmatrix} x_1 & y & y & \cdots & y & y+0 \\ z & x_2 & y & \cdots & y & y+0 \\ z & z & x_3 & \cdots & y & y+0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ z & z & z & \cdots & x_{n-1} & y+0 \\ z & z & z & \cdots & z & y+x_n-y \end{vmatrix} = \begin{vmatrix} x_1 & y & y & \cdots & y & y \\ z & x_2 & y & \cdots & y & y \\ z & z & x_3 & \cdots & y & y \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ z & z & z & \cdots & x_{n-1} & y \\ z & z & z & \cdots & z & y \end{vmatrix} + \begin{vmatrix} x_1 & y & y & \cdots & y & 0 \\ z & x_2 & y & \cdots & y & 0 \\ z & z & x_3 & \cdots & y & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ z & z & z & \cdots & x_{n-1} & 0 \\ z & z & z & \cdots & z & x_n-y \end{vmatrix} \\
 &= \begin{vmatrix} x_1-z & 0 & 0 & \cdots & 0 & 0 \\ 0 & x_2-z & 0 & \cdots & 0 & 0 \\ 0 & 0 & x_3-z & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x_{n-1}-z & 0 \\ z & z & z & \cdots & z & y \end{vmatrix} + (x_n-y) D_{n-1} = y \prod_{i=1}^{n-1} (x_i-z) + (x_n-y) D_{n-1}. \quad (2)
 \end{aligned}$$

将原行列式转置后, 同理可得

$$\begin{aligned}
 D_n = D_n^T &= \begin{vmatrix} x_1 & z & z & \cdots & z & z+0 \\ y & x_2 & z & \cdots & z & z+0 \\ y & y & x_3 & \cdots & z & z+0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ y & y & y & \cdots & x_{n-1} & z+0 \\ y & y & y & \cdots & y & z+x_n-z \end{vmatrix} = \begin{vmatrix} x_1 & z & z & \cdots & z & z \\ y & x_2 & z & \cdots & z & z \\ y & y & x_3 & \cdots & z & z \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ y & y & y & \cdots & x_{n-1} & z \\ y & y & y & \cdots & y & z \end{vmatrix} + \begin{vmatrix} x_1 & z & z & \cdots & z & 0 \\ y & x_2 & z & \cdots & z & 0 \\ y & y & x_3 & \cdots & z & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ y & y & y & \cdots & x_{n-1} & 0 \\ y & y & y & \cdots & y & x_n-z \end{vmatrix} \\
 &= \begin{vmatrix} x_1-y & 0 & 0 & \cdots & 0 & 0 \\ 0 & x_2-y & 0 & \cdots & 0 & 0 \\ 0 & 0 & x_3-y & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x_{n-1}-y & 0 \\ y & y & y & \cdots & y & z \end{vmatrix} + (x_n-z) D_{n-1}^T = z \prod_{i=1}^{n-1} (x_i-y) + (x_n-z) D_{n-1}. \quad (3)
 \end{aligned}$$

若 $z \neq y$, 则联立(2)(3)式, 解得

$$D_n = \frac{1}{z-y} \left[z \prod_{i=1}^n (x_i-y) - y \prod_{i=1}^n (x_i-z) \right];$$

若 $z = y$, 则由(2)式递推可得

$$\begin{aligned}
 D_n &= y \prod_{i=1}^{n-1} (x_i-y) + (x_n-y) D_{n-1} \\
 &= y \prod_{i=1}^{n-1} (x_i-y) + (x_n-y) \left(y \prod_{i=1}^{n-2} (x_i-y) + (x_{n-1}-y) D_{n-2} \right) \\
 &= y \prod_{j \neq n} (x_j-y) + y \prod_{j \neq n-1} (x_j-y) + (x_n-y)(x_{n-1}-y) D_{n-2} \\
 &= \cdots = y \sum_{i=1}^n \prod_{j \neq i} (x_j-y) + \prod_{i=1}^n (x_i-y) D_0 \\
 &= y \sum_{i=1}^n \prod_{j \neq i} (x_j-y) + \prod_{i=1}^n (x_i-y).
 \end{aligned}$$

解法二 (大拆分法): 令 $f(x) \triangleq \begin{vmatrix} x_1+x & y+x & \cdots & y+x \\ z+x & x_2+x & \cdots & y+x \\ \vdots & \vdots & & \vdots \\ z+x & z+x & \cdots & x_n+x \end{vmatrix}$, 则 $f(x)$ 一定是线性函数, 从而设 $f(x) = ax + b$. 注

意到

$$f(-z) = \begin{vmatrix} x_1-z & y-z & \cdots & y-z \\ 0 & x_2-z & \cdots & y-z \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & x_n-z \end{vmatrix} = \prod_{i=1}^n (x_i - z), \quad f(-y) = \begin{vmatrix} x_1-y & 0 & \cdots & 0 \\ z-y & x_2-y & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ z-y & z-y & \cdots & x_n-y \end{vmatrix} = \prod_{i=1}^n (x_i - y).$$

当 $y \neq z$ 时, 将上式代入 $f(x) = ax + b$ (即线性函数 $f(x)$ 过两点 $(-y, f(-y)), (-z, f(-z))$, 再利用两点式) 解得

$$f(x) = \frac{f(-z) - f(-y)}{-z - (-y)}(x + y) + f(-y) = \frac{\prod_{i=1}^n (x_i - z) - \prod_{i=1}^n (x_i - y)}{y - z}(x + y) + \prod_{i=1}^n (x_i - y).$$

从而此时就有

$$D_n = f(0) = \frac{y \prod_{i=1}^n (x_i - z) - z \prod_{i=1}^n (x_i - y)}{y - z}. \quad (4)$$

当 $y = z$ 时, 将 D_n 看作关于 y 的连续函数, 记为 $g(y) = D_n$, 则此时由 g 的连续性及(4)式和 L'Hospital 法则可得

$$\begin{aligned} D_n = g(z) &= \lim_{y \rightarrow z} g(y) = \lim_{y \rightarrow z} \frac{y \prod_{i=1}^n (x_i - z) - z \prod_{i=1}^n (x_i - y)}{y - z} \\ &= \lim_{y \rightarrow z} \frac{\prod_{i=1}^n (x_i - z) + y \sum_{i=1}^n \prod_{j \neq i} (x_j - y)}{1} = \prod_{i=1}^n (x_i - z) + z \sum_{i=1}^n \prod_{j \neq i} (x_j - z). \end{aligned}$$

□

例题 0.2

(1) 计算

$$|B| = \begin{vmatrix} 2a_1 & a_1 + a_2 & \cdots & a_1 + a_n \\ a_2 + a_1 & 2a_2 & \cdots & a_2 + a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n + a_1 & a_n + a_2 & \cdots & 2a_n \end{vmatrix}.$$

(2) 求下列 n 阶行列式的值:

$$|A| = \begin{vmatrix} 0 & a_1 + a_2 & \cdots & a_1 + a_{n-1} & a_1 + a_n \\ a_2 + a_1 & 0 & \cdots & a_2 + a_{n-1} & a_2 + a_n \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n-1} + a_1 & a_{n-1} + a_2 & \cdots & 0 & a_{n-1} + a_n \\ a_n + a_1 & a_n + a_2 & \cdots & a_n + a_{n-1} & 0 \end{vmatrix}.$$

笔记 Show/Hide Note

第(2)问解法一中不仅使用了升阶法还使用了分块”爪”型行列式的计算方法. 观察到各行各列有不同的公共项, 因此可以利用升阶法将各行各列的公共项消去.

注 因为第(2)问中, 当 $a_i \neq 0 (i = 1, 2, \cdots, n)$ 时, 最后的结果不含 a_i 的分式结构, 所以当存在 $a_i = 0$, 其中 $i \in 1, 2, \cdots, ns$ 时, 根据行列式 (可以看作多元多项式函数) 的连续性可知, 此时最后的结果就是将 a_i 中相应为零的值代

入当 $a_i \neq 0 (i = 1, 2, \dots, n)$ 时的结果中. 因此我们直接不妨设 $a_i \neq 0 (i = 1, 2, \dots, n)$, 只需考虑这一种情况即可.

解 Show/Hide Solution

(1) 注意到

$$B = \begin{pmatrix} 2a_1 & a_1 + a_2 & \cdots & a_1 + a_n \\ a_2 + a_1 & 2a_2 & \cdots & a_2 + a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n + a_1 & a_n + a_2 & \cdots & 2a_n \end{pmatrix} = \begin{pmatrix} 1 & a_1 \\ 1 & a_2 \\ \vdots & \vdots \\ 1 & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}.$$

由 Cauchy-Binet 公式可知

$$|B| = \begin{cases} 0, & n \geq 3, \\ -(a_1 - a_2)^2, & n = 2, \\ 2a_1, & n = 1. \end{cases}$$

(2) (i) 当 $a_i \neq 0 (1 \leq i \leq n)$ 时, **解法一 (升阶法)**:

$$\begin{aligned} |A| &\xrightarrow{\text{升阶}} \begin{vmatrix} 1 & -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 0 & 0 & a_1 + a_2 & \cdots & a_1 + a_{n-1} & a_1 + a_n \\ 0 & a_2 + a_1 & 0 & \cdots & a_2 + a_{n-1} & a_2 + a_n \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n-1} + a_1 & a_{n-1} + a_2 & \cdots & 0 & a_{n-1} + a_n \\ 0 & a_n + a_1 & a_n + a_2 & \cdots & a_n + a_{n-1} & 0 \end{vmatrix} \\ &\xrightarrow[r_1+r_i]{i=1,2,\dots,n+1} \begin{vmatrix} 1 & -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 1 & -a_1 & a_1 & \cdots & a_1 & a_1 \\ 1 & a_2 & -a_2 & \cdots & a_2 & a_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & a_{n-1} & a_{n-1} & \cdots & -a_{n-1} & a_{n-1} \\ 1 & a_n & a_n & \cdots & a_n & -a_n \end{vmatrix} \xrightarrow{\text{升阶}} \begin{vmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ -a_1 & 1 & -a_1 & a_1 & \cdots & a_1 & a_1 \\ -a_2 & 1 & a_2 & -a_2 & \cdots & a_2 & a_2 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ -a_{n-1} & 1 & a_{n-1} & a_{n-1} & \cdots & -a_{n-1} & a_{n-1} \\ -a_n & 1 & a_n & a_n & \cdots & a_n & -a_n \end{vmatrix} \\ &\xrightarrow[j_1+j_i]{i=1,3,4,\dots,n+2} \begin{vmatrix} 1 & 0 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 1 & -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ -a_1 & 1 & -2a_1 & 0 & \cdots & 0 & 0 \\ -a_2 & 1 & 0 & -2a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ -a_{n-1} & 1 & 0 & 0 & \cdots & -2a_{n-1} & 0 \\ -a_n & 1 & 0 & 0 & \cdots & 0 & -2a_n \end{vmatrix} \\ &\xrightarrow[-\frac{1}{2}j_i+j_1]{-\frac{1}{2}j_i+j_1} \begin{vmatrix} 1-\frac{n}{2} & \frac{S}{2} & 1 & 1 & \cdots & 1 & 1 \\ \frac{T}{2} & 1-\frac{n}{2} & -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 0 & 0 & -2a_1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & -2a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -2a_{n-1} & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & -2a_n \end{vmatrix}. \end{aligned}$$

其中 $S = a_1 + a_2 + \cdots + a_n, T = \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}$. 注意到上述行列式是分块上三角行列式, 从而可得

$$\begin{aligned} |A| &= (-2)^n \prod_{i=1}^n a_i \cdot \frac{(n-2)^2 - ST}{4} = (-2)^{n-2} \prod_{i=1}^n a_i [(n-2)^2 - (\sum_{i=1}^n a_i)(\sum_{i=1}^n \frac{1}{a_i})] \\ &= (-2)^{n-2} \prod_{i=1}^n a_i (n-2)^2 - (-2)^{n-2} \sum_{i=1}^n a_i \sum_{j=1}^n \prod_{k \neq j} a_k. \end{aligned}$$

解法二 (矩阵和的行列式计算公式)(不推荐使用!):

$$\text{设 } B = \begin{pmatrix} 2a_1 & a_1 + a_2 & \cdots & a_1 + a_n \\ a_2 + a_1 & 2a_2 & \cdots & a_2 + a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n + a_1 & a_n + a_2 & \cdots & 2a_n \end{pmatrix}, C = \begin{pmatrix} -2a_1 & & & \\ & -2a_2 & & \\ & & \ddots & \\ & & & -2a_n \end{pmatrix}, \text{ 则 } |A| = |B + C|.$$

从而利用矩阵和的行列式计算公式的结论得到

$$|A| = |B| + |C| + \sum_{1 \leq k \leq n-1} \left(\sum_{\substack{1 \leq i_1 < i_2 < \cdots < i_k \leq n \\ 1 \leq j_1 < j_2 < \cdots < j_k \leq n}} B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} \widehat{C} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} \right) \quad (5)$$

其中 $\widehat{C} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 是 $C \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 的代数余子式.

我们先来计算 $B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}, k = 1, 2, \cdots, n$. 拆分 $B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 的第一列得到

$$\begin{aligned} B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} &= \begin{vmatrix} a_{i_1} + a_{j_1} & a_{i_1} + a_{j_2} & \cdots & a_{i_1} + a_{j_k} \\ a_{i_2} + a_{j_1} & a_{i_2} + a_{j_2} & \cdots & a_{i_2} + a_{j_k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_k} + a_{j_1} & a_{i_k} + a_{j_2} & \cdots & a_{i_k} + a_{j_k} \end{vmatrix} \\ &= \begin{vmatrix} a_{i_1} & a_{i_1} + a_{j_2} & \cdots & a_{i_1} + a_{j_k} \\ a_{i_2} & a_{i_2} + a_{j_2} & \cdots & a_{i_2} + a_{j_k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_k} & a_{i_k} + a_{j_2} & \cdots & a_{i_k} + a_{j_k} \end{vmatrix} + \begin{vmatrix} a_{j_1} & a_{i_1} + a_{j_2} & \cdots & a_{i_1} + a_{j_k} \\ a_{j_1} & a_{i_2} + a_{j_2} & \cdots & a_{i_2} + a_{j_k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j_1} & a_{i_k} + a_{j_2} & \cdots & a_{i_k} + a_{j_k} \end{vmatrix} \\ &= \begin{vmatrix} a_{i_1} & a_{j_2} & \cdots & a_{j_k} \\ a_{i_2} & a_{j_2} & \cdots & a_{j_k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_k} & a_{j_2} & \cdots & a_{j_k} \end{vmatrix} + \begin{vmatrix} a_{j_1} & a_{i_1} & \cdots & a_{i_1} \\ a_{j_1} & a_{i_2} & \cdots & a_{i_2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j_1} & a_{i_k} & \cdots & a_{i_k} \end{vmatrix} \end{aligned}$$

因此当 $k \geq 3$ 时, $B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} = 0$; 当 $k = 2$ 时, $B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} = B \begin{pmatrix} i_1 & i_2 \\ j_1 & j_2 \end{pmatrix} = \begin{vmatrix} a_{i_1} & a_{j_2} \\ a_{i_2} & a_{j_2} \end{vmatrix} + \begin{vmatrix} a_{j_1} & a_{i_1} \\ a_{j_1} & a_{i_2} \end{vmatrix} = (a_{i_1}a_{j_2} - a_{i_2}a_{j_2})(a_{i_2}a_{j_1} - a_{i_1}a_{j_1})$; 当 $k = 1$ 时, $B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} = B \begin{pmatrix} i_1 \\ j_1 \end{pmatrix} = a_{i_1} + a_{j_1}$.

又注意到 $|C|$ 只有主子式非零, 而其主子式 $C \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ i_1 & i_2 & \cdots & i_k \end{pmatrix} = (-2)^k a_{i_1} a_{i_2} \cdots a_{i_k}$. 于是当 $\exists m \in$

$\{1, 2, \dots, k\}$, 使得 $i_m \neq j_m$ 时, $\hat{C} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} = 0$; 当 $i_m = j_m, m = 1, 2, \dots, k$ 时, 有

$$\hat{C} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} = \hat{C} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ i_1 & i_2 & \cdots & i_k \end{pmatrix} = (-2)^{n-k} a_1 \cdots \hat{a}_{i_1} \cdots \hat{a}_{i_2} \cdots \hat{a}_{i_k} \cdots a_n.$$

故当 $n \geq 3$ 时, (5) 式可化为

$$\begin{aligned} |A| &= |B| + |C| + \sum_{1 \leq k \leq n-1} \left(\sum_{\substack{1 \leq i_1 < i_2 < \cdots < i_k \leq n \\ 1 \leq j_1 < j_2 < \cdots < j_k \leq n}} B \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} \hat{C} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} \right) \\ &= |C| + \sum_{\substack{1 \leq i_1 \leq n \\ 1 \leq j_1 \leq n}} B \begin{pmatrix} i_1 \\ j_1 \end{pmatrix} \hat{C} \begin{pmatrix} i_1 \\ j_1 \end{pmatrix} + \sum_{\substack{1 \leq i_1 < i_2 \leq n \\ 1 \leq j_1 < j_2 \leq n}} B \begin{pmatrix} i_1 & i_2 \\ j_1 & j_2 \end{pmatrix} \hat{C} \begin{pmatrix} i_1 & i_2 \\ j_1 & j_2 \end{pmatrix} \\ &= |C| + \sum_{1 \leq i_1 \leq n} B \begin{pmatrix} i_1 \\ i_1 \end{pmatrix} \hat{C} \begin{pmatrix} i_1 \\ i_1 \end{pmatrix} + \sum_{1 \leq i_1 < i_2 \leq n} B \begin{pmatrix} i_1 & i_2 \\ i_1 & i_2 \end{pmatrix} \hat{C} \begin{pmatrix} i_1 & i_2 \\ i_1 & i_2 \end{pmatrix} \\ &= |C| + \sum_{1 \leq i \leq n} B \begin{pmatrix} i \\ i \end{pmatrix} \hat{C} \begin{pmatrix} i \\ i \end{pmatrix} + \sum_{1 \leq i < j \leq n} B \begin{pmatrix} i & j \\ i & j \end{pmatrix} \hat{C} \begin{pmatrix} i & j \\ i & j \end{pmatrix} \\ &= (-2)^n a_1 a_2 \cdots a_n + \sum_{1 \leq i \leq n} 2a_i (-2)^{n-1} a_1 \cdots \hat{a}_i \cdots a_n + \sum_{1 \leq i < j \leq n} [(a_i a_j - a_j^2)(a_i a_j - a_i^2)(-2)^{n-2} a_1 \cdots \hat{a}_i \cdots \hat{a}_j \cdots a_n] \\ &= (-2)^n a_1 a_2 \cdots a_n - (-2)^n \sum_{1 \leq i \leq n} a_1 a_2 \cdots \cdots a_n + (-2)^{n-2} \sum_{1 \leq i < j \leq n} [-(a_i - a_j)^2 a_1 \cdots \hat{a}_i \cdots \hat{a}_j \cdots a_n] \\ &= (-2)^n a_1 a_2 \cdots a_n - (-2)^n n a_1 a_2 \cdots \cdots a_n - (-2)^{n-2} \sum_{1 \leq i < j \leq n} [(a_i - a_j)^2 a_1 \cdots \hat{a}_i \cdots \hat{a}_j \cdots a_n] \\ &= (-2)^n \prod_{i=1}^n a_i (1-n) - (-2)^{n-2} \prod_{i=1}^n a_i \sum_{1 \leq i < j \leq n} \frac{(a_i - a_j)^2}{a_i a_j} \\ &= (-2)^{n-2} \prod_{i=1}^n a_i [(n-2)^2 - (\sum_{i=1}^n a_i)(\sum_{i=1}^n \frac{1}{a_i})] \\ &= (-2)^n \prod_{i=1}^n a_i (1-n) - (-2)^{n-2} \prod_{i=1}^n a_i \sum_{1 \leq i < j \leq n} \frac{(a_i - a_j)^2}{a_i a_j} \\ &= (-2)^{n-2} \prod_{i=1}^n a_i \left[4 - 4n - \sum_{1 \leq i < j \leq n} \frac{(a_i - a_j)^2}{a_i a_j} \right] \\ &= (-2)^{n-2} \prod_{i=1}^n a_i \left[4 - 4n - \sum_{1 \leq i < j \leq n} \left(\frac{a_j}{a_i} + \frac{a_i}{a_j} - 2 \right) \right] \\ &= (-2)^{n-2} \prod_{i=1}^n a_i \left[4 - 4n - \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} \frac{a_i}{a_j} + \sum_{1 \leq i < j \leq n} 2 \right] \\ &= (-2)^{n-2} \prod_{i=1}^n a_i \left[4 - 4n - \left(\sum_{1 \leq i, j \leq n} \frac{a_i}{a_j} - \sum_{i=1}^n \frac{a_i}{a_i} \right) + \sum_{i=1}^{n-1} \sum_{j=i+1}^n 2 \right] \\ &= (-2)^{n-2} \prod_{i=1}^n a_i \left[4 - 4n - \left(\sum_{1 \leq i, j \leq n} \frac{a_i}{a_j} - n \right) + 2 \sum_{i=1}^{n-1} (n-i) \right] \\ &= (-2)^{n-2} \prod_{i=1}^n a_i \left[4 - 4n + n + n(n-1) - \sum_{i=1}^n \sum_{j=1}^n \frac{a_i}{a_j} \right] \end{aligned}$$

$$\begin{aligned}
&= (-2)^{n-2} \prod_{i=1}^n a_i \left[n^2 - 4n + 4 - \sum_{i=1}^n a_i \sum_{j=1}^n \frac{1}{a_j} \right] \\
&= (-2)^{n-2} \prod_{i=1}^n a_i \left[(n-2)^2 - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n \frac{1}{a_i} \right) \right] \\
&= (-2)^{n-2} \prod_{i=1}^n a_i (n-2)^2 - (-2)^{n-2} \sum_{i=1}^n a_i \sum_{j=1}^n \prod_{k \neq j} a_k.
\end{aligned}$$

解法三 (降价公式)(推荐使用!): 令 $A = \begin{pmatrix} a_1 & 1 \\ a_2 & 1 \\ \vdots & \vdots \\ a_n & 1 \end{pmatrix}$, $B = \begin{pmatrix} -2a_1 & & & \\ & -2a_2 & & \\ & & \ddots & \\ & & & -2a_n \end{pmatrix}$, 则

$$A = \begin{pmatrix} -2a_1 & & & \\ & -2a_2 & & \\ & & \ddots & \\ & & & -2a_n \end{pmatrix} + \begin{pmatrix} a_1 & 1 \\ a_2 & 1 \\ \vdots & \vdots \\ a_n & 1 \end{pmatrix} I_2^{-1} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \end{pmatrix} = B + \Lambda I_2^{-1} \Lambda'.$$

于是由降价公式 (打洞原理) 我们有

$$\begin{aligned}
|A| &= |I_2| |B + \Lambda I_2^{-1} \Lambda'| = \begin{vmatrix} I_2 & \Lambda' \\ \Lambda & B \end{vmatrix} = |B| |I_2 - \Lambda' B^{-1} \Lambda| \\
&= \begin{vmatrix} -2a_1 & & & \\ & -2a_2 & & \\ & & \ddots & \\ & & & -2a_n \end{vmatrix} \cdot \left| I_2 - \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \end{pmatrix} \begin{pmatrix} -\frac{1}{2a_1} & & & \\ & -\frac{1}{2a_2} & & \\ & & \ddots & \\ & & & -\frac{1}{2a_n} \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ a_2 & 1 \\ \vdots & \vdots \\ a_n & 1 \end{pmatrix} \right| \\
&= (-2)^n \prod_{i=1}^n a_i \left| I_2 - \begin{pmatrix} -\frac{1}{2a_1} & -\frac{1}{2a_2} & \cdots & -\frac{1}{2a_n} \\ \frac{1}{2} & \frac{1}{2} & \cdots & \frac{1}{2} \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ a_2 & 1 \\ \vdots & \vdots \\ a_n & 1 \end{pmatrix} \right| \\
&= (-2)^n \prod_{i=1}^n a_i \left| I_2 - \begin{pmatrix} -\frac{n}{2} & -\frac{1}{2} \sum_{i=1}^n \frac{1}{a_i} \\ -\frac{1}{2} \sum_{i=1}^n a_i & -\frac{n}{2} \end{pmatrix} \right| = (-2)^n \prod_{i=1}^n a_i \begin{vmatrix} \frac{n+2}{2} & \frac{1}{2} \sum_{i=1}^n \frac{1}{a_i} \\ \frac{1}{2} \sum_{i=1}^n a_i & \frac{n+2}{2} \end{vmatrix} \\
&= (-2)^{n-2} \prod_{i=1}^n a_i \left[(n+2)^2 - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n \frac{1}{a_i} \right) \right] \\
&= (-2)^{n-2} \prod_{i=1}^n a_i (n-2)^2 - (-2)^{n-2} \sum_{i=1}^n a_i \sum_{j=1}^n \prod_{k \neq j} a_k.
\end{aligned}$$

(ii) 当存在 $a_i = 0$, 其中 $i \in \{1, 2, \dots, n\}$ 时, 不妨设只有 $a_{i_1}, a_{i_2}, \dots, a_{i_m} = 0, i_1, i_2, \dots, i_m \in 1, 2, \dots, n$, 则可将 $|A|$ 看作关于 $a_{i_1}, a_{i_2}, \dots, a_{i_m}$ 连续的多元多项式函数 $g(a_{i_1}, a_{i_2}, \dots, a_{i_m})$, 于是由 g 的连续性可得

$$\begin{aligned}
g(0, 0, \dots, 0) &= \lim_{(a_{i_1}, a_{i_2}, \dots, a_{i_m}) \rightarrow (0, 0, \dots, 0)} g(a_{i_1}, a_{i_2}, \dots, a_{i_m}) \\
&= \lim_{(a_{i_1}, a_{i_2}, \dots, a_{i_m}) \rightarrow (0, 0, \dots, 0)} \left[(-2)^{n-2} \prod_{i=1}^n a_i (n-2)^2 - (-2)^{n-2} \sum_{i=1}^n a_i \sum_{j=1}^n \prod_{k \neq j} a_k \right] = 0.
\end{aligned}$$

即由行列式的连续性可知

$$|A| = (-2)^{n-2} \prod_{i=1}^n a_i (n-2)^2 - (-2)^{n-2} \sum_{i=1}^n a_i \sum_{j=1}^n \prod_{k \neq j} a_k.$$

对某些 a_i 为 0 时也成立.

□

结论 对角矩阵行列式的子式和余子式:

设 $|A| = \begin{vmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{vmatrix}$, 则其 k 阶子式 $A \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 除 k 阶主子式 $A \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ i_1 & i_2 & \cdots & i_k \end{pmatrix}$ 外都为

零, 其中 $k = 1, 2, \dots, n$.

记 $\hat{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 为 $A \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 的代数余子式 ($n-k$ 阶). 于是 $\hat{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix}$ 除 $\hat{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ i_1 & i_2 & \cdots & i_k \end{pmatrix}$ 外也都为零, 其中 $k = 1, 2, \dots, n$.

并且

$$A \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ i_1 & i_2 & \cdots & i_k \end{pmatrix} = a_{i_1} a_{i_2} \cdots a_{i_k},$$

$$\hat{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_k \\ i_1 & i_2 & \cdots & i_k \end{pmatrix} = a_1 \cdots \hat{a}_{i_1} \cdots \hat{a}_{i_2} \cdots \hat{a}_{i_k} \cdots a_n$$

其中 $k = 1, 2, \dots, n$.

命题 0.4 (Cauchy 行列式)

证明:

$$|A| = \begin{vmatrix} (a_1 + b_1)^{-1} & (a_1 + b_2)^{-1} & \cdots & (a_1 + b_n)^{-1} \\ (a_2 + b_1)^{-1} & (a_2 + b_2)^{-1} & \cdots & (a_2 + b_n)^{-1} \\ \vdots & \vdots & \ddots & \vdots \\ (a_n + b_1)^{-1} & (a_n + b_2)^{-1} & \cdots & (a_n + b_n)^{-1} \end{vmatrix} = \frac{\prod_{1 \leq i < j \leq n} (a_j - a_i)(b_j - b_i)}{\prod_{1 \leq i < j \leq n} (a_i + b_j)}.$$

笔记 Show/Hide Note

需要记忆 Cauchy 行列式的计算方法.

1. 分式分母有公共部分可以作差, 得到的分子会变得相对简便.

2. 行列式内行列做加减一般都是加减同一行 (或列). 但是在循环行列式中, 我们一般采取相邻两行 (或列) 相加减的方法.

证明 Show/Hide Proof

$$|A| = \begin{vmatrix} \frac{1}{a_1 + b_1} & \frac{1}{a_1 + b_2} & \cdots & \frac{1}{a_1 + b_n} \\ \frac{1}{a_2 + b_1} & \frac{1}{a_2 + b_2} & \cdots & \frac{1}{a_2 + b_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{a_n + b_1} & \frac{1}{a_n + b_2} & \cdots & \frac{1}{a_n + b_n} \end{vmatrix}$$

$$\begin{aligned}
& \frac{-j_n+j_i}{i=n-1, \dots, 1} \left| \begin{array}{cccc} \frac{b_n-b_1}{(a_1+b_1)(a_1+b_n)} & \frac{b_n-b_2}{(a_1+b_2)(a_1+b_n)} & \cdots & \frac{b_n-b_{n-1}}{(a_1+b_{n-1})(a_1+b_n)} & \frac{1}{a_1+b_n} \\ \frac{b_n-b_1}{(a_2+b_1)(a_2+b_n)} & \frac{b_n-b_2}{(a_2+b_2)(a_2+b_n)} & \cdots & \frac{b_n-b_{n-1}}{(a_1+b_{n-1})(a_2+b_n)} & \frac{1}{a_2+b_n} \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{b_n-b_1}{(a_n+b_1)(a_n+b_n)} & \frac{b_n-b_2}{(a_n+b_2)(a_n+b_n)} & \cdots & \frac{b_n-b_{n-1}}{(a_1+b_{n-1})(a_n+b_n)} & \frac{1}{a_n+b_n} \end{array} \right| \\
&= \frac{\prod_{i=1}^{n-1} (b_n-b_i)}{\prod_{j=1}^n (a_j+b_n)} \left| \begin{array}{cccc} \frac{1}{a_1+b_1} & \frac{1}{a_1+b_2} & \cdots & \frac{1}{a_1+b_{n-1}} & 1 \\ \frac{1}{a_2+b_1} & \frac{1}{a_2+b_2} & \cdots & \frac{1}{a_2+b_{n-1}} & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{1}{a_n+b_1} & \frac{1}{a_n+b_2} & \cdots & \frac{1}{a_n+b_{n-1}} & 1 \end{array} \right| \\
& \frac{-r_n+r_i}{i=n-1, \dots, 1} \frac{\prod_{i=1}^{n-1} (b_n-b_i)}{\prod_{j=1}^n (a_j+b_n)} \left| \begin{array}{cccc} \frac{a_n-a_1}{(a_1+b_1)(a_n+b_1)} & \frac{a_n-a_1}{(a_1+b_2)(a_n+b_2)} & \cdots & \frac{a_n-a_1}{(a_1+b_{n-1})(a_n+b_{n-1})} & 0 \\ \frac{a_n-a_1}{(a_2+b_1)(a_n+b_1)} & \frac{a_n-a_1}{(a_2+b_2)(a_n+b_2)} & \cdots & \frac{a_n-a_1}{(a_2+b_{n-1})(a_n+b_{n-1})} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{a_n-a_{n-1}}{(a_{n-1}+b_1)(a_n+b_1)} & \frac{a_n-a_{n-1}}{(a_{n-1}+b_2)(a_n+b_2)} & \cdots & \frac{a_n-a_{n-1}}{(a_{n-1}+b_{n-1})(a_n+b_{n-1})} & 0 \\ \frac{1}{a_n+b_1} & \frac{1}{a_n+b_2} & \cdots & \frac{1}{a_n+b_{n-1}} & 1 \end{array} \right| \\
&= \frac{\prod_{i=1}^{n-1} (b_n-b_i)}{\prod_{j=1}^n (a_j+b_n)} \cdot \frac{\prod_{i=1}^{n-1} (a_n-a_i)}{\prod_{k=1}^{n-1} (a_n+b_k)} \left| \begin{array}{cccc} \frac{1}{a_1+b_1} & \frac{1}{a_1+b_2} & \cdots & \frac{1}{a_1+b_{n-1}} & 0 \\ \frac{1}{a_2+b_1} & \frac{1}{a_2+b_2} & \cdots & \frac{1}{a_2+b_{n-1}} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{1}{a_{n-1}+b_1} & \frac{1}{a_{n-1}+b_2} & \cdots & \frac{1}{a_{n-1}+b_{n-1}} & 0 \\ 1 & 1 & \cdots & 1 & 1 \end{array} \right| \\
& \frac{\text{按最后一列展开}}{\prod_{j=1}^n (a_j+b_n)} \frac{\prod_{i=1}^{n-1} (b_n-b_i)(a_n-a_i)}{\prod_{k=1}^{n-1} (a_n+b_k)} \left| \begin{array}{cccc} \frac{1}{a_1+b_1} & \frac{1}{a_1+b_2} & \cdots & \frac{1}{a_1+b_{n-1}} \\ \frac{1}{a_2+b_1} & \frac{1}{a_2+b_2} & \cdots & \frac{1}{a_2+b_{n-1}} \\ \vdots & \vdots & & \vdots \\ \frac{1}{a_{n-1}+b_1} & \frac{1}{a_{n-1}+b_2} & \cdots & \frac{1}{a_{n-1}+b_{n-1}} \end{array} \right| \\
&= \frac{\prod_{i=1}^{n-1} (b_n-b_i)(a_n-a_i)}{\prod_{j=1}^n (a_j+b_n) \prod_{k=1}^{n-1} (a_n+b_k)} \cdot D_{n-1}.
\end{aligned}$$

不断递推下去即得

$$\begin{aligned}
D_n &= \frac{\prod_{i=1}^{n-1} (b_n-b_i)(a_n-a_i)}{\prod_{j=1}^n (a_j+b_n) \prod_{k=1}^{n-1} (a_n+b_k)} \cdot D_{n-1} = \frac{\prod_{i=1}^{n-1} (b_n-b_i)(a_n-a_i)}{\prod_{j=1}^n (a_j+b_n) \prod_{k=1}^{n-1} (a_n+b_k)} \cdot \frac{\prod_{i=1}^{n-2} (b_{n-1}-b_i)(a_{n-1}-a_i)}{\prod_{j=1}^{n-1} (a_j+b_{n-1}) \prod_{k=1}^{n-2} (a_{n-1}+b_k)} \cdot D_{n-2} \\
&= \cdots = \frac{\prod_{i=1}^{n-1} (b_n-b_i)(a_n-a_i)}{\prod_{j=1}^n (a_j+b_n) \prod_{k=1}^{n-1} (a_n+b_k)} \cdot \frac{\prod_{i=1}^{n-2} (b_{n-1}-b_i)(a_{n-1}-a_i)}{\prod_{j=1}^{n-1} (a_j+b_{n-1}) \prod_{k=1}^{n-2} (a_{n-1}+b_k)} \cdots \frac{\prod_{i=1}^2 (b_3-b_i)(a_3-a_i)}{\prod_{j=1}^3 (a_j+b_3) \prod_{k=1}^2 (a_3+b_k)} \cdot D_2
\end{aligned}$$

$$\begin{aligned}
&= \frac{\prod_{i=1}^{n-1} (b_n - b_i)(a_n - a_i)}{\prod_{j=1}^n (a_j + b_n) \prod_{k=1}^{n-1} (a_n + b_k)} \cdot \frac{\prod_{i=1}^{n-2} (b_{n-1} - b_i)(a_{n-1} - a_i)}{\prod_{j=1}^{n-1} (a_j + b_{n-1}) \prod_{k=1}^{n-2} (a_{n-1} + b_k)} \cdots \frac{\prod_{i=1}^2 (b_3 - b_i)(a_3 - a_i)}{\prod_{j=1}^3 (a_j + b_3) \prod_{k=1}^2 (a_3 + b_k)} \cdot \frac{(b_2 - b_1)(a_2 - a_1)}{\prod_{j=1}^2 (a_j + b_2)(a_2 + b_1)} \cdot D_1 \\
&= \frac{\prod_{i=1}^{n-1} (b_n - b_i)(a_n - a_i)}{\prod_{j=1}^n (a_j + b_n) \prod_{k=1}^{n-1} (a_n + b_k)} \cdot \frac{\prod_{i=1}^{n-2} (b_{n-1} - b_i)(a_{n-1} - a_i)}{\prod_{j=1}^{n-1} (a_j + b_{n-1}) \prod_{k=1}^{n-2} (a_{n-1} + b_k)} \cdots \frac{\prod_{i=1}^2 (b_3 - b_i)(a_3 - a_i)}{\prod_{j=1}^3 (a_j + b_3) \prod_{k=1}^2 (a_3 + b_k)} \cdot \frac{(b_2 - b_1)(a_2 - a_1)}{\prod_{j=1}^2 (a_j + b_2)(a_2 + b_1)} \cdot \frac{1}{a_1 + b_1} \\
&= \frac{\prod_{1 \leq i < j \leq n} (a_j - a_i)(b_j - b_i)}{\prod_{1 \leq i < j \leq n} (a_i + b_j) \prod_{1 \leq j < i \leq n} (a_i + b_j)} = \frac{\prod_{1 \leq i < j \leq n} (a_j - a_i)(b_j - b_i)}{\prod_{1 \leq i < j \leq m} (a_i + b_j)}.
\end{aligned}$$

□

例题 0.3 设 n 阶行列式

$$A_n = \begin{vmatrix} a_0 + a_1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ a_1 & a_1 + a_2 & a_2 & 0 & \cdots & 0 & 0 \\ 0 & a_2 & a_2 + a_3 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{n-1} & a_{n-1} + a_n \end{vmatrix},$$

求证:

$$A_n = a_0 a_1 \cdots a_n \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_n} \right).$$

 **笔记** Show/Hide Note用**数学归纳法**证明与行列式有关的结论.练习0.1和练习0.2都可同理使用用数学归纳法证明(对阶数 n 进行归纳即可).**证明** Show/Hide Proof(数学归纳法) 对阶数 n 进行归纳. 当 $n = 1, 2$ 时, 结论显然成立. 假设阶数小于 n 结论成立.现证明 n 阶的情形. 注意到

$$A_n = \begin{vmatrix} a_0 + a_1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ a_1 & a_1 + a_2 & a_2 & 0 & \cdots & 0 & 0 \\ 0 & a_2 & a_2 + a_3 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{n-1} & a_{n-1} + a_n \end{vmatrix} = (a_{n-1} + a_n) A_{n-1} - a_{n-1}^2 A_{n-2}.$$

将归纳假设代入上面的式子中得

$$\begin{aligned}
A_n &= (a_{n-1} + a_n) A_{n-1} - a_{n-1}^2 A_{n-2} \\
&= (a_{n-1} + a_n) a_0 a_1 \cdots a_{n-1} \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_{n-1}} \right) - a_{n-1}^2 a_0 a_1 \cdots a_{n-2} \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_{n-2}} \right) \\
&= a_0 a_1 \cdots a_n \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_{n-1}} \right) + a_0 a_1 \cdots a_{n-2} a_{n-1}^2 \frac{1}{a_{n-1}} \\
&= a_0 a_1 \cdots a_{n-1} \left[a_n \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_{n-1}} \right) + 1 \right] \\
&= a_0 a_1 \cdots a_{n-1} a_n \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_{n-1}} + \frac{1}{a_n} \right).
\end{aligned}$$

故由数学归纳法可知, 结论对任意正整数 n 都成立.

□

例题 0.4 设 $n(n > 2)$ 阶行列式 $|A|$ 的所有元素或为 1 或为 -1, 求证: $|A|$ 的绝对值小于等于 $\frac{2}{3}n!$.

解 Show/Hide Solution

对阶数 n 进行归纳. 当 $n = 3$ 时, 将 $|A|$ 的第一列元素为 -1 的行都乘以 -1 , 再将 $|A|$ 的第一行元素为 1 的列都乘以 -1 , $|A|$ 的绝对值不改变.

$$\text{因此不妨设 } |A| = \begin{vmatrix} 1 & -1 & -1 \\ 1 & a_0 & b_0 \\ 1 & c_0 & d_0 \end{vmatrix}, \text{ 其中 } a_0, b_0, c_0, d_0 = 1 \text{ 或 } -1.$$

从而

$$|A| = \begin{vmatrix} 1 & -1 & -1 \\ 1 & a_0 & b_0 \\ 1 & c_0 & d_0 \end{vmatrix} \xrightarrow[i=2,3]{j_1+j_i} \begin{vmatrix} 1 & 0 & 0 \\ 1 & a & b \\ 1 & c & d \end{vmatrix}, \text{ 其中 } a, b, c, d = 0 \text{ 或 } 2.$$

于是

$$\text{abs}(|A|) = \text{abs} \left(\begin{vmatrix} 1 & 0 & 0 \\ 1 & a & b \\ 1 & c & d \end{vmatrix} \right) = \text{abs}(ad - bc) \leq 4 = \frac{2}{3} \cdot 3!$$

假设 $n-1$ 阶时结论成立, 现证 n 阶的情形. 将 $|A|$ 按第一行展开得

$$|A| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n}, \text{ 其中 } a_{1i} = 1 \text{ 或 } -1 (i = 1, 2, \cdots, n).$$

从而由归纳假设可得

$$\begin{aligned} \text{abs}(|A|) &= \text{abs}(a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n}) \leq \text{abs}(A_{11}) + \text{abs}(A_{12}) + \cdots + \text{abs}(A_{1n}) \\ &\leq \frac{2}{3}(n-1)! + \frac{2}{3}(n-1)! + \cdots + \frac{2}{3}(n-1)! \\ &= n \cdot \frac{2}{3}(n-1)! = \frac{2}{3}n!. \end{aligned}$$

故由数学归纳法可知结论对任意正整数都成立. □

命题 0.5 (行列式的求导运算)

对 $\forall n \in \mathbb{N}$, 设 $f_{ij}(t) (i, j = 1, 2, \cdots, n)$ 都是可微函数,

$$F(t) = \begin{vmatrix} f_{11}(t) & f_{12}(t) & \cdots & f_{1n}(t) \\ f_{21}(t) & f_{22}(t) & \cdots & f_{2n}(t) \\ \vdots & \vdots & & \vdots \\ f_{n1}(t) & f_{n2}(t) & \cdots & f_{nn}(t) \end{vmatrix}$$

求证: $\frac{d}{dt}F(t) = \sum_{j=1}^n F_j(t)$, 其中

$$F_j(t) = \begin{vmatrix} f_{11}(t) & f_{12}(t) & \cdots & \frac{d}{dt}f_{1j}(t) & \cdots & f_{1n}(t) \\ f_{21}(t) & f_{22}(t) & \cdots & \frac{d}{dt}f_{2j}(t) & \cdots & f_{2n}(t) \\ \vdots & \vdots & & \vdots & & \vdots \\ f_{n1}(t) & f_{n2}(t) & \cdots & \frac{d}{dt}f_{nj}(t) & \cdots & f_{nn}(t) \end{vmatrix}$$

笔记 Show/Hide Note

证法三实际上就是把 F 视作复合函数

$$\begin{aligned} F &= F(a_{1j}, \cdots, a_{nj}, A_{1j}, \cdots, A_{nj}) = F(a_{1j}, \cdots, a_{nj}, A_{1j}((a_{kl})_{k=1, \cdots, n; l \neq j}), \cdots, A_{nj}((a_{kl})_{k=1, \cdots, n; l \neq j})) \\ &= F(a_{1j}(t), \cdots, a_{nj}(t), A_{1j}((a_{kl}(t))_{k=1, \cdots, n; l \neq j}), \cdots, A_{nj}((a_{kl}(t))_{k=1, \cdots, n; l \neq j})). \end{aligned}$$

然后利用链式法则求导.

证明 Show/Hide Proof

证法一 (数学归纳法): 对阶数 n 进行归纳. 当 $n = 1$ 时结论显然成立. 假设 $n - 1$ 阶时结论成立, 现证 n 阶的情形.

将 $F(t)$ 按第一列展开得

$$F(t) = f_{11}(t)A_{11}(t) + f_{21}(t)A_{21}(t) + \cdots + f_{n1}(t)A_{n1}(t).$$

其中 $A_{i1}(t)$ 是元素 $f_{i1}(t)$ 的代数余子式. ($i = 1, 2, \cdots, n$)

从而由归纳假设可得

$$A'_{i1}(t) = \frac{d}{dt}A_{i1}(t) = \sum_{k=2}^n A_{i1}^k(t), i = 1, 2, \cdots, n.$$

$$\text{其中 } A_{i1}^k(t) = \begin{vmatrix} f_{12}(t) & \cdots & \frac{d}{dt}f_{1k}(t) & \cdots & f_{1n}(t) \\ \vdots & & \vdots & & \vdots \\ f_{i-1,2}(t) & \cdots & \frac{d}{dt}f_{i-1,k}(t) & \cdots & f_{i-1,n}(t) \\ f_{i+1,2}(t) & \cdots & \frac{d}{dt}f_{i+1,k}(t) & \cdots & f_{i+1,n}(t) \\ \vdots & & \vdots & & \vdots \\ f_{n2}(t) & \cdots & \frac{d}{dt}f_{nk}(t) & \cdots & f_{nn}(t) \end{vmatrix}, k = 2, 3, \cdots, n.$$

于是, 我们就有

$$\begin{aligned} \frac{d}{dt}F(t) &= \frac{d}{dt} [f_{11}(t)A_{11}(t) + f_{21}(t)A_{21}(t) + \cdots + f_{n1}(t)A_{n1}(t)] \\ &= f'_{11}(t)A_{11}(t) + f'_{21}(t)A_{21}(t) + \cdots + f'_{n1}(t)A_{n1}(t) + f_{11}(t)A'_{11}(t) + f_{21}(t)A'_{21}(t) + \cdots + f_{n1}(t)A'_{n1}(t) \\ &= \sum_{i=1}^n f'_{i1}(t)A_{i1}(t) + f_{11}(t) \sum_{k=2}^n A_{11}^k(t) + f_{21}(t) \sum_{k=2}^n A_{21}^k(t) + \cdots + f_{n1}(t) \sum_{k=2}^n A_{n1}^k(t) \\ &= \sum_{i=1}^n f'_{i1}(t)A_{i1}(t) + \sum_{i=1}^n \left(f_{i1}(t) \sum_{k=2}^n A_{i1}^k(t) \right) \\ &= \sum_{i=1}^n f'_{i1}(t)A_{i1}(t) + \sum_{i=1}^n f_{i1}(t) (A_{i1}^2 + A_{i1}^3 + \cdots + A_{i1}^n) \\ &= \sum_{i=1}^n f'_{i1}(t)A_{i1}(t) + \sum_{i=1}^n f_{i1}(t)A_{i1}^2 + \sum_{i=1}^n f_{i1}(t)A_{i1}^3 + \cdots + \sum_{i=1}^n f_{i1}(t)A_{i1}^n \\ &= F_1(t) + F_2(t) + F_3(t) + \cdots + F_n(t) \\ &= \sum_{j=1}^n F_j(t). \end{aligned}$$

故由数学归纳法可知结论对任意正整数都成立.

证法二 (行列式的组合定义): 由行列式的组合定义可得

$$F(t) = \sum_{1 \leq k_1, k_2, \cdots, k_n \leq n} (-1)^{\tau(k_1 k_2 \cdots k_n)} f_{k_1 1}(t) f_{k_2 2}(t) \cdots f_{k_n n}(t).$$

因此

$$\begin{aligned} \frac{d}{dt}F(t) &= \sum_{1 \leq k_1, k_2, \cdots, k_n \leq n} (-1)^{\tau(k_1 k_2 \cdots k_n)} f'_{k_{11}}(t) f_{k_{22}}(t) \cdots f_{k_{nn}}(t) \\ &\quad + \sum_{1 \leq k_1, k_2, \cdots, k_n \leq n} (-1)^{\tau(k_1 k_2 \cdots k_n)} f_{k_{11}}(t) f'_{k_{22}}(t) \cdots f_{k_{nn}}(t) \end{aligned}$$

$$\begin{aligned}
& + \cdots + \sum_{1 \leq k_1, k_2, \dots, k_n \leq n} (-1)^{\tau(k_1 k_2 \dots k_n)} f_{k_{11}}(t) f_{k_{22}}(t) \cdots f'_{k_{nn}}(t) \\
& = F_1(t) + F_2(t) + \cdots + F_n(t).
\end{aligned}$$

证法三: 记 $A_{ij}(t)$ 为元素 $f_{ij}(t)$ 的代数余子式. 固定第 j 列, 按第 j 列展开行列式得

$$F(t) = \sum_{i=1}^n f_{ij}(t) A_{ij}(t). \quad (6)$$

将 F 视为关于 $f_{1j}, \dots, f_{nj}$ 以及 $A_{1j}, \dots, A_{nj}$ 的复合函数, 则(6)式两边分别关于 A_{ij}, f_{ij} 得

$$\frac{\partial F}{\partial A_{ij}} = f_{ij}, \quad \frac{\partial F}{\partial f_{ij}} = A_{ij}, \quad i = 1, 2, \dots, n. \quad (7)$$

再由链式法则得

$$\frac{dF}{dt} = \sum_{i=1}^n \frac{\partial F}{\partial f_{ij}} \frac{df_{ij}}{dt} + \sum_{i=1}^n \frac{\partial F}{\partial A_{ij}} \frac{dA_{ij}}{dt} = \sum_{i=1}^n A_{ij} \frac{df_{ij}}{dt} + \sum_{i=1}^n f_{ij} \frac{dA_{ij}}{dt}. \quad (8)$$

由于 A_{ij} 仅依赖于除第 j 列以外的元素 $f_{kl} (l \neq j)$, 应用链式法则得

$$\frac{dA_{ij}}{dt} = \sum_{l \neq j} \sum_{k=1}^n \frac{\partial A_{ij}}{\partial f_{kl}} \cdot \frac{df_{kl}}{dt}, \quad i = 1, 2, \dots, n. \quad (9)$$

将(9)代入(8)的第二项, 得

$$\sum_{i=1}^n f_{ij} \frac{dA_{ij}}{dt} = \sum_{l \neq j} \sum_{k=1}^n \left(\sum_{i=1}^n f_{ij} \frac{\partial A_{ij}}{\partial f_{kl}} \right) \cdot \frac{df_{kl}}{dt}. \quad (10)$$

当 $l \neq j$ 时, 考虑 F 关于 f_{kl} 的偏导数. 由(6)(7)式且 $f_{ij} (i = 1, \dots, n)$ 中不含 f_{kl} , 有

$$A_{kl} = \frac{\partial F}{\partial f_{kl}} = \sum_{i=1}^n \frac{\partial}{\partial f_{kl}} (f_{ij} A_{ij}) = \sum_{i=1}^n f_{ij} \frac{\partial A_{ij}}{\partial f_{kl}}, \quad 1 \leq k \leq n, l \in \{1, 2, \dots, n\} \setminus \{j\}. \quad (11)$$

将(11)式代入(10)式, 得到

$$\sum_{i=1}^n f_{ij} \frac{dA_{ij}}{dt} = \sum_{l \neq j} \sum_{k=1}^n A_{kl} \frac{df_{kl}}{dt}. \quad (12)$$

将(12)式代回(8)式得

$$\frac{dF}{dt} = \sum_{i=1}^n A_{ij} \frac{df_{ij}}{dt} + \sum_{l \neq j} \sum_{k=1}^n A_{kl} \frac{df_{kl}}{dt} = \sum_{k=1}^n \sum_{l=1}^n A_{kl} \frac{df_{kl}}{dt}.$$

注意 $\sum_{k=1}^n A_{kl} \frac{df_{kl}}{dt}$ 正是将行列式 F 的第 l 列元素替换为其导数后再按第 l 列展开所得的行列式, 因此

$$\frac{dF}{dt} = \sum_{l=1}^n F_l(t),$$

□